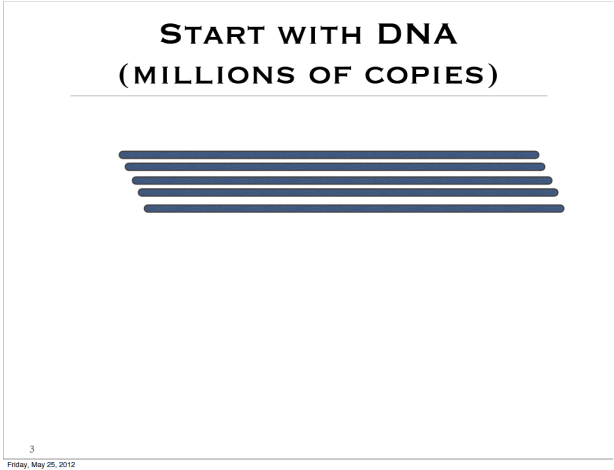
Biology | Technologies



https://www.youtube.com/watch?v=6TfYnsyo77o

BREAK IT

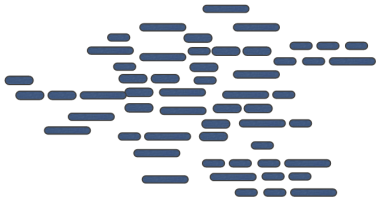


4

Friday, May 25, 2012

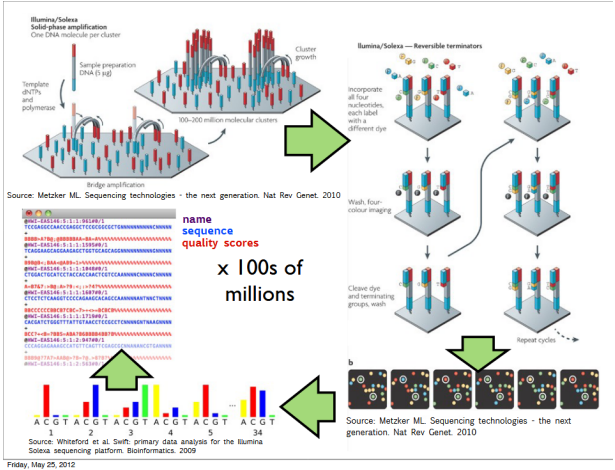
https://www.youtube.com/watch?v=6TfYnsyo77o

PUT IN SEQUENCER



https://www.youtube.com/watch?v=6TfYnsyo77o

Huge numbers of fragments can be sequenced simultaneously.

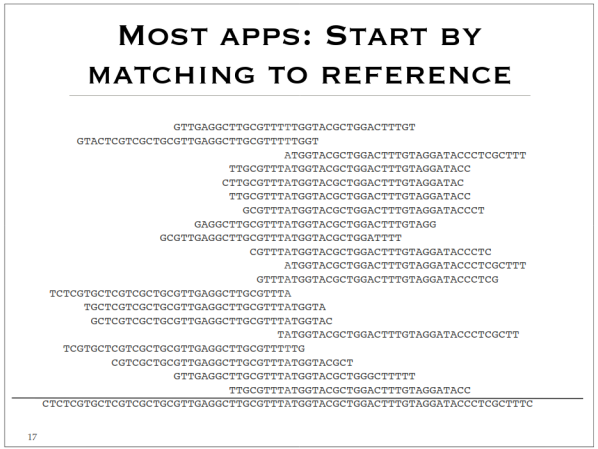


Friday, May 25, 2012

https://www.youtube.com/watch?v=6TfYnsyo77o

The sequenced part of each fragment (the reads) are stored in FASTQ files.

```
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+
@CCFFFEFFHHHHHJJJJIJJCGHEIIIJIJJJJ#1BFHIJJJJJJIJJIJJI#-;###-#-#-5?BFFFFEEEEEECCDDDDDDDDDDCCDDDDDDCCEED
@HWI-ST508:210:C0EDTACXX:1:1101:1895:1233 1:N:0:
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+
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+
CCCFFFFFHGHHHJJJJJJJIJJJJJJJIEIIIJFHGIIIIJJIJJJJJHIJJIJ#-;FGGIJIJHHFFDDEEDDCCDDDDCCDDDDDDCDDDDDDDDDDD
@HWI-ST508:210:C0EDTACXX:1:1101:1971:1236 1:N:0:
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+
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+
?@@DD?DEFBBFDHEHIGEDA@FH>C??BBBCB6B##########################################################################
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+
@@@DDA2?FHBHHEGEHIHGIGGHBFCGIEHGAEGGIIEGIIIIGHIGEHEGHIGIGBFHEHIEAHGHHFHEH;B@DEBDCDEEBCDDCCCCC@@CCCDCC
@HWI-ST508:210:C0EDTACXX:1:1101:1806:1245 1:N:0:
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+
CCCFFFFFHHHHGJJHJJJIJIJJJJJJJJJJJJJJJJJIIJJJJJJJJJJJJJJJJJGIJJJJIJJJHIJGIHHHGFFFEFFEDDDDDDDDDDEDD
```


MOST APPS: START BY MATCHING TO REFERENCE
GTTGAGGCTTGCGTTTTTGGTACGCTGGACTTTGT
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CTTGCGTTTATGGTACGCTGGACTTTGTAGGATAC
TTGCGTTTATGGTACGCTGGACTTTGTAGGATACC
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17

```
@HD     VN:1.0  SO:coordinate
@SQ     SN:chr20        LN:64444167
@PG     ID:TopHat       VN:2.0.14       CL:/srv/dna_tools/tophat/tophat -N 3 --read-edit-dist 5 --read-rea
lign-edit-dist 2 -i 50 -I 5000 --max-coverage-intron 5000 -M -o out /data/user446/mapping_tophat/index/chr
20 /data/user446/mapping_tophat/L6_18_GTGAAA_L007_R1_001.fastq
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C    BBDCCDDCCDDDDCDDDDDDCDCCCDBC?DDDDDDDDDDDDDDDDCCDCDDDDDDDDDDCCCCEDDDC?DDDDDDDDDDDDDDDDDDDDDDBDHFFFFDC@@
    AS:i:-15        XM:i:3  XO:i:0  XG:i:0  MD:Z:55C20C13A9 NM:i:3  NH:i:2  CC:Z:=  CP:i:55352714   HI:i:0
HWI-ST1145:74:C101DACXX:7:1114:2759:41961        16      chr20   193953  50      100M    *       0       0
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G    DCDDDDEDDDDDDDCDDDDDDDCCCDDDCDDDDDEEC>DFFFEJJJJJIGJJJJIHGBHHGJIJJJJJJGJJJIJJJJJJIHJJJJJJHHHHHFFFFFCCC
    AS:i:-16        XM:i:3  XO:i:0  XG:i:0  MD:Z:60G16T18T3 NM:i:3  NH:i:1
HWI-ST1145:74:C101DACXX:7:1204:14760:4030        16      chr20   270877  50      100M    *       0       0
       GGCTTTATTGGTAAAAAAGGAATAGCAGATTTAATCAGAAATTCCCACCTGGCCCAGCAGCACCAACCAGAAAGAAGGGAAGAAGACAGGAAAAAACCA
C    DDDDDDDDDCCDDDDDDDDDEEEEEEEFFFEFFEGHHHHFGDJJIHJJIJIJJJIIIIGGFJJIHIIIIJJJJJJIGHHFAHGFHJHFGGHFFDD@BB
    AS:i:-11        XM:i:2  XO:i:0  XG:i:0  MD:Z:0A85G13    NM:i:2  NH:i:1
HWI-ST1145:74:C101DACXX:7:1210:11167:8699        0       chr20   271218  50      50M4700N50M     *       0
       0       GTGGCTCTTCCACAGGAATGTTGAGGATGACATCCATGTCTGGGGTGCACTTGGGTCTCCGAAGCAGAACATCCTCAAATATGACCTCTCG
accepted_hits.sam
```

We will discuss three major applications of next-generation sequencing.



Bulk

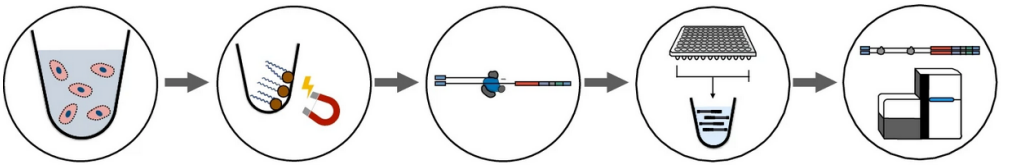


Single Cell



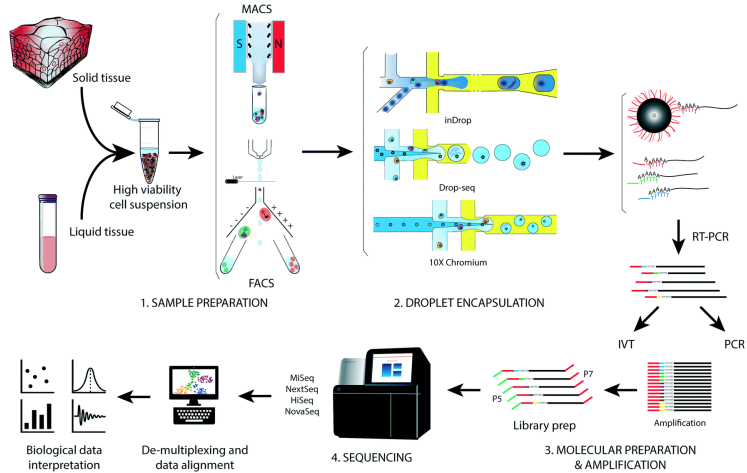
Spatial

Bulk RNA-seq pools all mRNAs from a tissue sample.

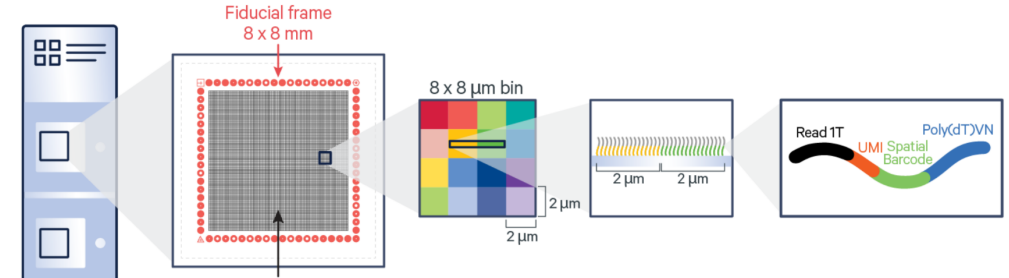


Janjic et al. (2022)

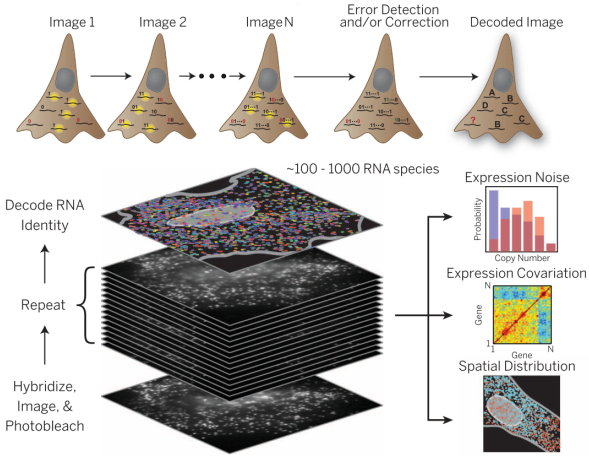
Single-cell RNA-seq uniquely labels mRNAs within individual cells.



Salomon et al. (2019)

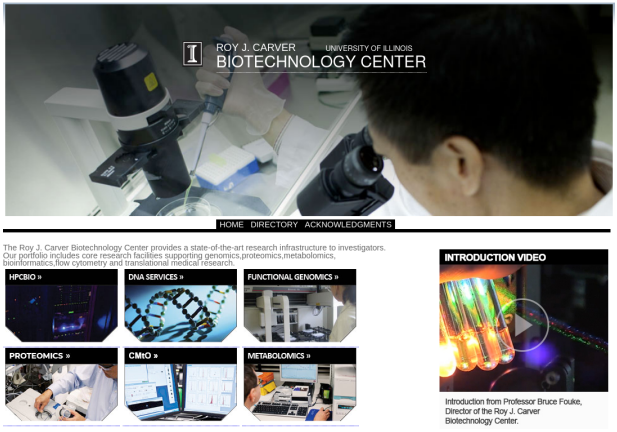

Fiducial frame
8 x 8 mm
8 x 8 µm bin
2 µm
2 µm
2 µm
2 µm
Read 1T
UMI
Spatial Barcode
Poly(dT)VN
Visium HD Slide, 6.5 mm
Capture Area with continuous lawn of oligos, 6.5 x 6.5 mm
Grid of 2 x 2 µm barcoded squares, binned to 8 x 8 µm
Oligo with Spatial Barcode

Sequencing is not the only high-throughput transcriptomic measurement technology.

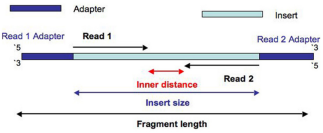


Chen at al. (2015)

Each technology requires a huge amount of specialized experimental design and data preprocessing.

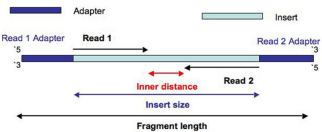


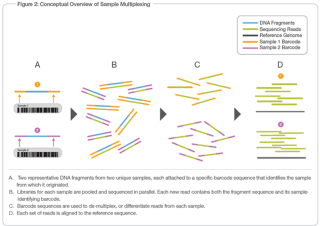
Examples of necessary preprocessing for bulk RNA-seq.



Biostars, Illumina, FastQC, STAR, Bray et al. (2016), Liao et al. (2014)

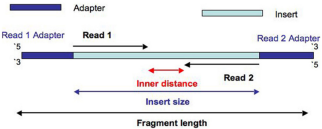
Examples of necessary preprocessing for bulk RNA-seq.

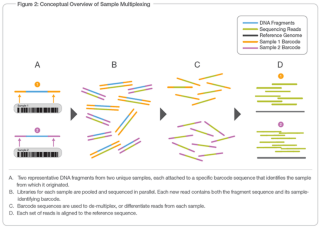


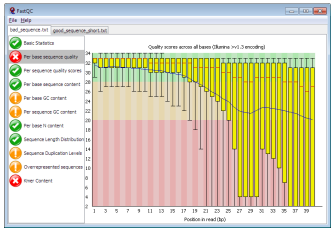


Biostars, Illumina, FastQC, STAR, Bray et al. (2016), Liao et al. (2014)

Examples of necessary preprocessing for bulk RNA-seq.

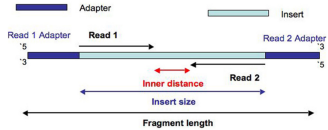


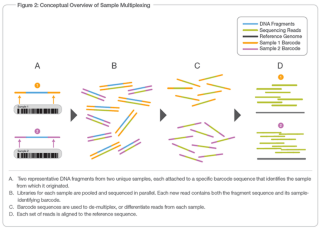


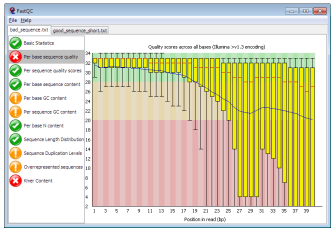


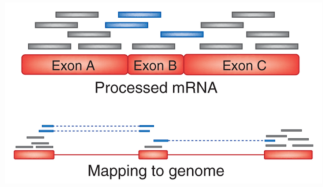
Biostars, Illumina, FastQC, STAR, Bray et al. (2016), Liao et al. (2014)

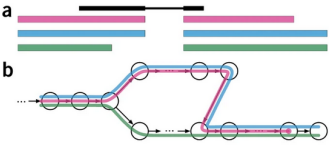
Examples of necessary preprocessing for bulk RNA-seq.





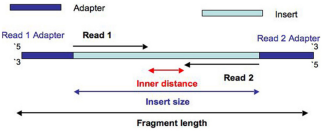


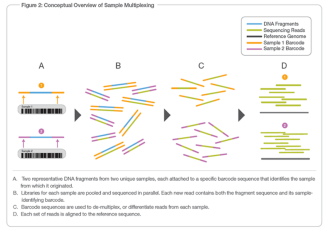


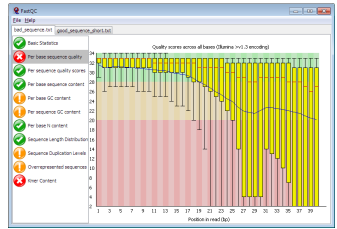


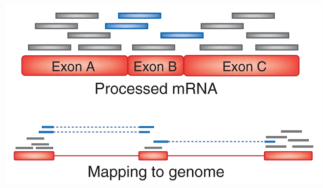
Biostars, Illumina, FastQC, STAR, Bray et al. (2016), Liao et al. (2014)

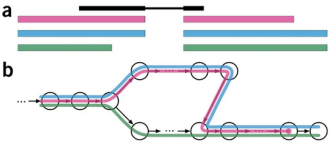
Examples of necessary preprocessing for bulk RNA-seq.

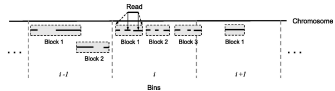












Biostars, Illumina, FastQC, STAR, Bray et al. (2016), Liao et al. (2014)

There are a huge number of competing techologies.

